

Hello Joshua,

You need to undertake the major revision of the thesis similarly to a paper revision for peer-review, of which you have good experience.

Please prepare a file with the reply, and a new version of the thesis files, where you will mark in colour all the changes mentioned in the reply.

Start with extended Conclusion section, as was suggested by Prof. Lucarini. Describe there what you have developed and detected, and possible future work, in particular, on the variability of the land vegetation, which will be possible in future when the model data will be made public. This is a straightforward thing to do, and hopefully it will not take long.

I think you need to explain in the introduction that you consider not only bifurcational but also transitional tipping points, in particular, in the observed data for which there is no analytical framework developed. In this context, analysis of hurricanes makes sense, and moreover, your results do confirm that this approach works.

The criticism of the lack of scaling by Prof. Ashkenazy should be affirmed in your reply, with added explanation that we DO NOT claim the presence of scaling in the data, but rather use the scaling exponents as a SENSORS of changes that precede a tipping point. It is not a true scaling or power-law behaviour, as the data indeed is often short (i.e. asymptotic scale is simply impossible to investigate). However, it is a sensitive measure (as it is demonstrated in model and in the real data) of the changing dynamics in the time series. In fact, even the actual values of indicators are of less importance, because often people who work with EWS apply Gaussian filter and other types of detrending, which immediately reduce auto-correlations and make smaller values of the indicators. It is not the values, but rather the increasing trend in the indicators is what provides EWS - he can be referred to the multiple papers in Nature and PNAS by the team of Prof. Scheffer.

The appearing crossover in the data which initially has no such crossover in scaling curves actually is an indicator of EWS, which builds up longer-correlated signal, and this will eventually lead to a tipping point.

To address his criticism of the impact of periodicities, you can add analysis of the hurricane data with removed 12-hour tidal waves for comparison.

Hello Joshua - 3 July 2020

What Prof. Ashkenazy criticises as not a real potential well (as we are dealing with a time series rather than with a physical well where a particle can be trapped), can be named as an “abstraction” of a potential, or a “symbolic” potential, and you can address him to the papers by Kwasniok

<https://royalsocietypublishing.org/doi/full/10.1098/rsta.2011.0472>
and Diltevsen (this is a recent paper, but there many others before
- you can find more links)

<https://link.springer.com/article/10.1007/s00382-018-4519-2>

Please prepare the first iteration of the revision and send to Tobias and me soon.

Best

Valerie